ELECTRIC VEHICLE MARKET ANALYSIS

A Data-Driven Study Using Tableau

PROJECT REPORT

DETAIL	INFORMATION
Project Title	Electric Vehicle Market Analysis Using Tableau
Dataset	Electric Vehicle Market Classification Dataset (training dataset)
Tool Used	Tableau Desktop
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Course / Department	BCA
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Submission Date	2 NOV

Submitted as an academic data visualisation project.

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Fill in the blank cover-page details before submission. Add screenshots of your actual dashboard and story if your evaluator expects visual evidence.

1. EXECUTIVE SUMMARY

This project presents an exploratory analysis of an Electric Vehicle (EV) market dataset using Tableau Desktop. It organises market information into visualisations and an interactive dashboard so that users can examine sales distribution, manufacturer performance, regional patterns, vehicle categories, model-year trends, battery types and selected vehicle performance indicators.

The completed workbook contains six worksheets, four KPI views, an interactive dashboard and a Tableau Story consisting of seven scenes. Three calculated fields were created. Market Region and Model Year filters were added, and Tableau Performance Recording was used for performance testing.

No numerical market conclusions or performance benchmarks are invented in this report. Add verified values from the final workbook where required.

2. INTRODUCTION

Electric vehicles are an important part of the transition toward alternative transportation technologies. As the EV market expands, manufacturers, vehicle categories, battery technologies and customer preferences create large volumes of information that can be difficult to interpret in raw tabular form.

Data visualisation transforms information into understandable charts, indicators and dashboards. Tableau enables users to compare segments, identify patterns and explore data through interactive filters. This project applies Tableau to an EV market dataset.

3. PROBLEM DEFINITION AND OBJECTIVES

Problem definition: The dataset contains dimensions and measures such as manufacturer, market region, category, model year, battery type, sales units, price, driving range, efficiency, acceleration and customer rating. Reviewing these fields manually makes comparisons difficult. A consolidated visual analysis is required.

Primary objective: To analyse the available EV market data using Tableau and communicate the analysis through visualisations, KPIs, an interactive dashboard and a multi-scene story.

4. DATASET OVERVIEW

The project uses **ev_market_train.csv** from the Electric Vehicle Market Classification Dataset associated with Skill Wallet. The file was connected to Tableau Desktop. The data source was reported as containing approximately **200,000 rows and 37 fields**. A separate test CSV was also available, but this analysis was built using the training file.

Examples of fields include Annual Sales Units, Market Region, Manufacturer, Category, Model Year, Battery Type, Efficiency Wh Per Km, Customer Rating, Price USD, Range Km, Vehicle Id and Acceleration 0 100 Sec.

5. TOOLS AND TECHNOLOGIES

Tableau Desktop: data connection, worksheets, calculated fields, KPI views, dashboard, story and performance recording.

CSV: source data format. **GitHub:** intended repository for sharing the packaged workbook and report. **Skill Wallet:** submission platform for the repository URL.

6. PROJECT METHODOLOGY

The workflow consisted of: (1) connecting the training CSV to Tableau; (2) reviewing dimensions, measures and identifiers; (3) creating six worksheets; (4) creating four KPI views; (5) defining three calculated fields; (6) assembling the worksheets and KPIs into a dashboard; (7) creating a seven-scene Tableau Story; and (8) testing interactions with Tableau Performance Recording and saving the workbook.

7. DATA PREPARATION AND UNDERSTANDING

The source CSV was connected directly to Tableau. Dimensions such as Manufacturer, Market Region, Category, Model Year and Battery Type were used to segment the data. Measures such as Annual Sales Units, Price USD, Range Km and

Customer Rating were used for aggregation and comparison.

This report does not claim that a separate cleaning pipeline, database extraction, missing-value treatment or duplicate-removal operation was performed, because those steps were not documented. Before submission, check null values, duplicate identifiers, inconsistent categories and data types, and describe only the cleaning steps actually completed.

8. VISUALISATION DESIGN

Six worksheets were developed to represent different aspects of EV sales and performance.

9. KEY PERFORMANCE INDICATORS (KPIs)

Four KPI views were created and arranged in one row at the top of the dashboard to provide a concise summary of the selected data.

10. CALCULATED FIELDS

Sales Contribution %

```
SUM([Annual Sales Units]) / TOTAL(SUM([Annual Sales Units]))
```

Calculates each displayed mark's share of total sales in the view. The TOTAL() table calculation depends on the view dimensions and Compute Using configuration.

Average Sales per Vehicle

```
SUM([Annual Sales Units]) / COUNTD([Vehicle Id])
```

Divides aggregated sales units by the distinct count of Vehicle Id. Its interpretation depends on the source data grain and the meaning of the identifier.

EV Customer Rating Status

```
IF AVG([Customer Rating]) >= 4 THEN "High Rated"
ELSEIF AVG([Customer Rating]) >= 3 THEN "Moderately Rated"
ELSE "Low Rated"
END
```

Classifies the aggregated rating into three bands. It was used on Colour in the EV Performance Analysis worksheet.

11. DASHBOARD DESIGN AND INTERACTIVITY

The dashboard is named **EV Market Performance Dashboard**. It uses Automatic sizing and arranges the six worksheets in a 3-by-2 layout. Four KPI views are positioned in one row. The title area reads: **ELECTRIC VEHICLE MARKET — PERFORMANCE & INSIGHTS**, with the subtitle *A Data-Driven Analysis of the Global EV Market*.

Two filters were added: **Market Region**, to focus on a selected region, and **Model Year**, to focus on selected model years. The creator reported that the filters were tested. For final quality assurance, confirm that both filters affect all intended worksheets and all four KPI values.

12. TABLEAU STORY

A Tableau Story was created with seven story points, guiding viewers from individual analysis views to the consolidated dashboard. The story was played through and all scenes were reported to appear correctly.

13. PERFORMANCE TESTING

Tableau's Performance Recording feature was used. The workflow involved starting Performance Recording, interacting with the dashboard and story, and stopping the recording to open the performance summary workbook.

The purpose is to inspect the relative cost of workbook operations and identify potentially slow queries or rendering activity. No numerical load time or benchmark is stated because no measurements were supplied. Retain a screenshot of the performance summary if available.

14. FINDINGS AND INTERPRETATION

The workbook offers complementary perspectives: manufacturer and category charts support segment comparisons; the treemap provides a regional view; the line chart presents the model-year pattern; the packed bubble chart compares battery types; and the performance worksheet provides a focused view of selected vehicle performance information.

The KPI row gives a high-level summary, while dashboard filters support exploration by region and model year. Specific conclusions such as the highest-selling manufacturer or leading region should be added only after checking actual values in Tableau.

15. LIMITATIONS

The analysis uses the supplied training CSV; the test CSV was not documented as part of the workbook analysis.

Exact aggregated chart values were not provided for this report. Database extraction, cleaning actions and null/duplicate handling are not claimed unless actually performed.

Average Sales per Vehicle depends on the dataset's record grain and the definitions of Vehicle Id and Annual Sales Units. Performance benchmarks were not provided.

16. CONCLUSION AND FUTURE SCOPE

The project demonstrates how Tableau can turn a multi-field CSV dataset into an organised analytical workbook. Six worksheets, four KPI views, three calculated fields, an interactive dashboard with two filters and a seven-scene Story were developed.

Future enhancements include adding verified numerical insights, documenting data validation and cleaning, adding further filters or map-based analysis where appropriate, and recording measured performance observations.

17. PROJECT DELIVERABLES

Recommended GitHub repository structure:

```
EV-Market-Analysis-Project/
EV_Market_Analysis_Project.twbx
EV_Market_Analysis_Project_Report.pdf
README.md
screenshots/
dashboard.png
story.png
performance_recording.png
```

Before uploading, verify that the packaged workbook opens correctly. Do not include credentials, personal information or unrelated files. If dataset redistribution is restricted or the file is too large, provide source attribution and access instructions instead of uploading it without permission.

WORKSHEET INVENTORY

No.	Worksheet	Chart type	Purpose
1	EV Sales by Manufacturer	Horizontal bar	Compare aggregated annual sales units by manufacturer.
2	EV Sales by Region	Treemap	Show relative regional contribution to the selected sales measure.
3	EV Sales by Category	Horizontal bar	Compare sales across EV categories.
4	EV Sales by Model Year	Line	Show year-wise pattern in aggregated EV sales.
5	EV Sales by Battery Type	Packed bubble	Compare battery types using bubble size.

No.	Worksheet	Chart type	Purpose
6	EV Performance Analysis	Vertical bar	Compare selected performance measure(s); rating status used for colour.

The six worksheets use five distinct chart types: horizontal bar, treemap, line, packed bubble and vertical bar. Horizontal bar charts are used in two worksheets.

KPI INVENTORY

KPI	Tableau aggregation	Meaning
Total EV Sales	SUM(Annual Sales Units)	Total sales units in selected records.
Average EV Price	AVG(Price USD)	Average vehicle price in USD.
Average Driving Range	AVG(Range Km)	Average driving range in kilometres.
Average Customer Rating	AVG(Customer Rating)	Average customer rating in selected records.

STORY SCENE INVENTORY

Scene	Story point title	View
1	Manufacturer Sales Analysis	EV Sales by Manufacturer
2	Regional EV Sales Analysis	EV Sales by Region
3	Category-wise EV Sales Analysis	EV Sales by Category
4	Year-wise EV Sales Trend	EV Sales by Model Year
5	Battery Type Sales Analysis	EV Sales by Battery Type
6	EV Performance Comparison	EV Performance Analysis
7	Final EV Market Dashboard	EV Market Performance Dashboard

END OF REPORT